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An Efficient CFO Estimation Algorithm for the Downlink of 3GPP-LTE

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Abstract—Synchronization is of fundamental importance for wireless cellular systems, especially for orthogonal frequency division multiplexing (OFDM) based systems. In this paper, we investigate carrier frequency offset (CFO) estimation in the initial synchronization for the downlink of 3GPP-long term evolution (3GPP-LTE). Firstly, three traditional CFO estimation schemes including the primary synchronization channel (P-SCH) signals (PSS) correlation based scheme, the cyclic prefix (CP) based scheme, and the multiple signal classification (MUSIC) based scheme, are respectively applied. Simulation results show that the PSS correlation based scheme can achieve better performance than the other two. Furthermore, we extend the traditional PSS correlation based scheme, and propose a more efficient CFO estimation algorithm. Simulation results show that, compared to the traditional estimator, the proposed estimator has better estimation performance in terms of normalized mean square error (MSE), and achieves a gain of around 2 dB over both additive white Gaussian noise (AWGN) channels and frequency selective fading channels.

Index Terms—3GPP-LTE downlink, OFDM, synchronization, CFO estimation, PSS correlation, MUSIC algorithm, mean square error.

I. INTRODUCTION

Orthogonal frequency division multiplexing (OFDM) has been adopted by many wireless communication standards for its advantages of effectively reducing the detrimental multipath effect and providing flexible resource allocation in frequency selective channels. 3GPP-long term evolution [1] (3GPP-LTE) has chosen orthogonal frequency division multiple access (OFDMA) as the air interface of downlink [2]. Because OFDMA combines the technologies of OFDM [3] and frequency division multiple access (FDMA), it offers several favorable features such as high spectral efficiency and simple receiver architecture, etc. However, OFDM/OFDMA is sensitive to carrier frequency offsets (CFOs) and timing offsets. To maintain the orthogonality of subcarriers, CFOs and timing errors must be estimated and adequately compensated for. Much intensive research effort has been devoted to this issue in recent years. Morelli *et al.* gave a comprehensive review on the synchronization techniques for OFDM/OFDMA systems [4].

In 3GPP-LTE systems, downlink synchronization is the fundamental task in cell search [5] when a user equipment (UE) tries to connect to a base station (BS), i.e., an evolved Node B (eNB) in terms of the 3GPP-LTE [1]. LTE provides a primary synchronization channel (P-SCH) and a secondary synchronization channel (S-SCH) to help a UE to detect

TABLE I
OVERVIEW OF 3GPP-LTE DOWNLINK SYNCHRONIZATION PROCEDURE.

Procedure	Purpose
1. Frequency spectrum scan	Find the carrier frequency and bandwidth of the closest cell.
2. P-SCH detection	Obtain symbol timing, 5 ms frame timing, and physical layer ID.
3. CFO estimation	Compensate for the CFO, and achieve frequency synchronization.
4. Channel estimation	For equalization.
5. S-SCH detection	Obtain 10 ms frame timing and cell ID.

and synchronize eNBs. By successfully detecting both P-SCH and S-SCH, the UE can obtain OFDM symbol timing and frame timing, and get the basic system information, such as cell identity (ID), downlink channel bandwidth and cyclic prefix (CP) type, etc. Tab. I shows a clear overview of the LTE downlink synchronization.

In this paper, the CFO estimation for the downlink of 3GPP-LTE is investigated. Several schemes in [4] have been investigated for the CFO estimation in OFDM/OFDMA systems. The methods of CFO estimation can be classified into two groups, i.e., the training symbol based (that is, data-aided) approaches such as [6]–[8], and blind estimation such as [9]–[11]. Moose in [6] proposed a pilot based CFO estimation scheme by detecting the phase shifts between several successive identical pilot blocks. Schmidl and Cox in [7] presented a robust synchronization scheme for OFDM using one unique OFDM symbol which has a repetition format within half a symbol period, and the acquisition is achieved in two separate steps through the use of a two-symbol training sequence. Morelli and Mengali in [8] proposed an improved CFO estimation scheme which is based on the best linear unbiased estimation (BLUE) principle. A maximum likelihood (ML) CFO estimation was originally proposed in [9] based on CP. Liu and Tureli in [10] presented a multiple signal classification (MUSIC) scheme which is based on subspace spanned by virtual subcarriers. For LTE systems, several CFO estimation schemes in [14] [15] have already been presented. In [14], the proposed CFO estimation scheme adopted the ML method for PSS correlation. Xu in [15] proposed a robust hierarchical scheme in which the CP based lagged auto-correlation is employed to estimate CFO and the PSS based correlation is used to achieve a more accurate symbol and frame timing.

This paper firstly reviews three CFO estimation schemes for the downlink of 3GPP-LTE, i.e., the PSS correlation based scheme, the CP based scheme, and the MUSIC scheme, respectively. The performance of three CFO estimation schemes is also provided. Then an extended CFO estimation algorithm based on the PSS correlation scheme is proposed. The receiver detects the transmitted PSS sequence by implementing the cross-correlation between the three local PSS sequences and the received PSS samples. Then, the proposed estimator divides the PSS correlation sequence into four parts, and get six pairwise combinations. The CFO can be obtained by calculating the phase difference of each pairwise combination. The final CFO estimation is the linear combination of six CFO estimates. The proposed algorithm extends the idea in the traditional scheme, and can make better use of the good correlation property of the PSS sequence. Simulation results will show that, compared to the traditional estimator, the proposed estimator has better estimation performance in terms of normalized mean square error (MSE), and achieves a around 2 dB gain over both additive white Gaussian noise (AWGN) channels and frequency selective fading channels.

The rest of this paper is organized as follows. In Section II, the PSS in the downlink of 3GPP LTE and the system model are described. Several traditional CFO estimation schemes are reviewed and compared in Section III. In Section IV, the proposed algorithm is presented and evaluated. Section V shows the physical layer simulation results, and compares the performance of the proposed scheme to that of the traditional one over the different channels. Finally, the conclusions are drawn in Section VI.

Notations: Italic letters denote scalars. Bold lower-case and bold upper-case letters denote vectors and matrices, respectively. The superscripts $(\cdot)^*$, $(\cdot)^T$, $(\cdot)^H$ denote complex conjugate, transpose, and Hermitian transpose, respectively. The notation $\mathbf{x} \sim \mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ means that $\mathbf{x}$ is a complex Gaussian random vector with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. $\mathbf{I}$ denotes the identity matrix and $\text{diag}(\mathbf{x})$ stands for the diagonal matrix with elements of the vector $\mathbf{x}$ on its main diagonal.

II. SYSTEM DESCRIPTION

A. Primary Synchronization Channel Signal (PSS)

As specified in [1], the time duration of a radio frames in LTE downlink is $T_{\text{frame}} = 10$ ms. Each frame consists of 10 subframes of equal length while each subframe contains two consecutive slots of time duration $T_{\text{slot}} = 0.5$ ms. The granularity for the user allocation is in terms of resource block (RB). Each RB is a time-frequency grid of 12 subcarriers over $N_{\text{sc}} = 7$ OFDM symbols for the normal CP length or $N_{\text{sc}} = 6$ for the extended CP length. In 3GPP-LTE [1] downlink system, for all transmission bandwidth, two dedicated synchronization channels P-SCH and S-SCH are located at the centered 1.08 MHz frequency band, corresponding to 72 subcarriers. The LTE radio frame structure in the frequency division duplex (FDD) mode is shown in Fig. 1.

3GPP-LTE adopts one of the three length-62 Zadoff-Chu (ZC) sequences [13] as the PSS, and one ZC sequence

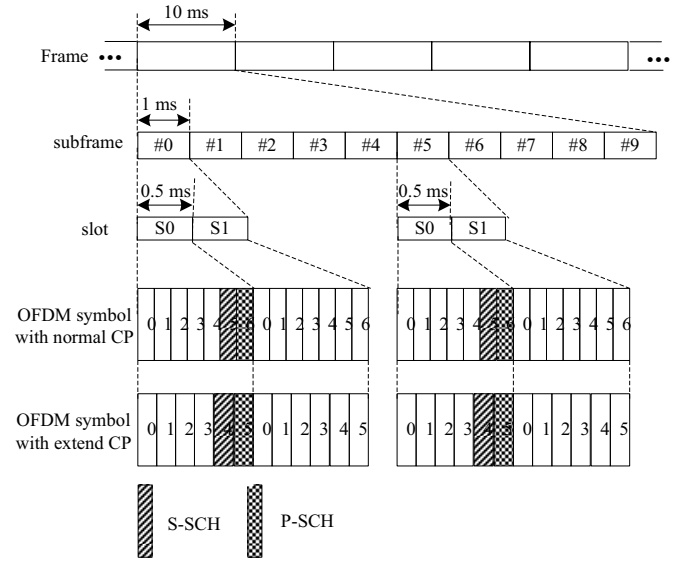


Fig. 1. Radio Frame Structures in LTE FDD downlink.

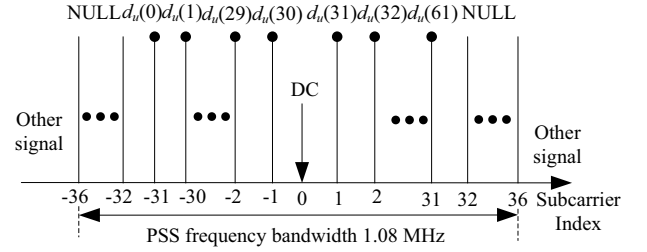


Fig. 2. Mapping of ZC sequence (1) to the subcarriers in 3GPP-LTE downlink.

is transmitted over P-SCH, which is located at both the first subframe and the sixth subframe, as shown in Fig. 1. The ZC sequence has the property of constant amplitude zero autocorrelation (CAZAC) [13], and the sequence is given by

$$d_u(k) = \begin{cases} \exp(-j \frac{\pi u n(n+1)}{63}), & k = 0, 1, \dots, 30 \\ \exp(-j \frac{\pi u (n+1)(n+2)}{63}), & k = 31, 32, \dots, 61 \end{cases} \quad (1)$$

where u is called ZC root index, and $u \in \{25, 29, 34\}$ for LTE systems. Fig. 2 describes how the ZC sequence (1) is mapped to the subcarriers. As shown in Fig. 2, there are 62 active subcarriers used around the DC subcarrier. Meanwhile, 5 null subcarriers at each end of P-SCH are reserved as the guard bandwidth. Note that the PSS sequence is uniquely determined by ZC root index u .

B. System Model

Consider the downlink of a 3GPP-LTE system with bandwidth B Hz and N subcarriers. The transmitted PSS $s_u(n)$ in

the time domain is generated by taking the N -point inverse discrete Fourier transform (IDFT) of the PSS sequence $d_u(k)$, that is

$$\mathbf{s}_u := [s_u(0), s_u(1), \dots, s_u(N-1)]^T = \mathbf{W}_P \mathbf{d}_u, \quad (2)$$

where $\mathbf{d}_u := [d_u(0), d_u(1), \dots, d_u(61)]^T$, u is randomly chosen from the set $\{25, 29, 34\}$, and $\mathbf{W}_P$ is a submatrix of the $N \times N$ IDFT matrix $\mathbf{W}$. The 62-length PSS sequence is modulated at the subcarriers whose index are in the set $\{-31, -32, \dots, -1, 1, 2, \dots, 31\}$, as shown in Fig. 2, hence $\mathbf{W}_P := [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{31}, \mathbf{w}_{N-31}, \mathbf{w}_{N-30}, \dots, \mathbf{w}_{N-1}]$, where $\mathbf{w}_p$ is the p^{th} column of $\mathbf{W}$.

With the presence of CFO and noise, and after removing the CP, the received time-domain complex baseband OFDM samples $r_{\text{pss}}(n)$ are given by

$$r_{\text{pss}}(n) = e^{j\frac{2\pi n\epsilon}{N}} \sum_{l=0}^{L-1} h_l s_u((n-l) \bmod N) + v(n), \quad n = 0, 1, \dots, N-1, \quad (3)$$

where h_l ($l = 0, 1, \dots, L-1$) is the overall channel impulse response (CIR), L is the maximum channel delay spread. ϵ denotes the CFO normalized to the subcarrier spacing. $v(n)$ is the i.i.d complex additive white Gaussian noise and $v(n) \sim \mathcal{CN}(0, \sigma_n^2)$.

Using matrix-vector representation, the received time-domain PSS samples (3) are rewritten as

$$\mathbf{r}_{\text{pss}} = \mathbf{\Gamma}(\epsilon) \mathbf{W}_P \mathbf{H} \mathbf{d}_u + \mathbf{v}, \quad (4)$$

where $\mathbf{r}_{\text{pss}} = [r_{\text{pss}}(0), r_{\text{pss}}(1), \dots, r_{\text{pss}}(N-1)]^T$, and the CFO matrix is $\mathbf{\Gamma}(\epsilon) = \text{diag}\{1, e^{j\frac{2\pi\epsilon}{N}}, e^{j\frac{4\pi\epsilon}{N}}, \dots, e^{j\frac{2\pi(N-1)\epsilon}{N}}\}$. $\mathbf{H} = \text{diag}\{H(0), H(1), \dots, H(N-1)\}$, where $H(i)$ is the channel frequency response, and $\mathbf{v} = [v(0), v(1), \dots, v(N-1)]^T$ is a zero-mean Gaussian vector which is subjected to $\mathbf{v} \sim \mathcal{CN}(0, \sigma_n^2 \mathbf{I}_N)$.

Based on the CAZAC property of PSS sequence, the root index u of the transmitted PSS sequence can be detected by taking the correlation between the received PSS samples (4) and the local PSS sequences $\mathbf{s}_f$, ($f \in \{25, 29, 34\}$). Define the correlation term as

$$y_f = \mathbf{s}_f^H \mathbf{r}_{\text{pss}} = \sum_{n=0}^{N-1} s_f^*(n) r_{\text{pss}}(n), \quad f \in \{25, 29, 34\}, \quad (5)$$

where $\mathbf{s}_f = \mathbf{W}_P \mathbf{d}_f$. Hence, in order to get the transmitted ZC sequence root index u , the PSS detection should be implemented by

$$\hat{u} = f^{\text{opt}} = \arg \max_f y_f. \quad (6)$$

Once the ZC sequence root index $\hat{u}$ is obtained, the optimal local PSS sequence $\mathbf{s}_{\hat{u}}$ can be chosen to estimate CFO in the following Sections.

III. THE TRADITIONAL CFO ESTIMATION SCHEMES

In this section, three CFO estimation schemes [6]-[12] including the pilot aided scheme, the CP based scheme, and MUSIC based scheme, are respectively reviewed and applied to 3GPP-LTE systems. The performance of the three CFO estimation schemes for 3GPP-LTE downlink over AWGN channels and frequency selective fading channels is also provided in this section through simulations.

A. The PSS Correlation Based Scheme

The PSS correlation based scheme borrows the idea of pilot aided scheme for OFDM systems and makes use of the CAZAC property of PSS sequence for 3GPP-LTE. The estimator is based on the correlation between the received PSS samples and the detected local PSS sequence $\mathbf{s}_{\hat{u}}$. Define the correlation term as

$$y = \mathbf{s}_{\hat{u}}^H \mathbf{r}_{\text{pss}} = \sum_{n=0}^{N-1} y(n), \quad (7)$$

where

$$\begin{aligned} y(n) &= s_{\hat{u}}^*(n) r_{\text{pss}}(n) \\ &= s_{\hat{u}}^*(n) \left(e^{j\frac{2\pi n\epsilon}{N}} \sum_{l=0}^{L-1} h_l s_u((n-l) \bmod N) + v(n) \right) \\ &= e^{j\frac{2\pi n\epsilon}{N}} h_0 s_{\hat{u}}^*(n) s_u(n) \\ &\quad + e^{j\frac{2\pi n\epsilon}{N}} \sum_{l \neq 0, l=1}^{L-1} h_l s_{\hat{u}}^*(n) s_u((n-l) \bmod N) \\ &\quad + s_{\hat{u}}^*(n) v(n), \quad n = 0, 1, \dots, N-1. \end{aligned} \quad (8)$$

Suppose that the PSS is detected correctly, that is $\hat{u} = u$, then (8) can be rewritten as

$$y(n) = e^{j\frac{2\pi n\epsilon}{N}} h_0 E_s + I(n) + \tilde{v}(n), \quad (9)$$

where $E_s := \|s_u(n)\|^2$, and E_s is constant for any n . $I(n)$, $\tilde{v}(n)$ denote the interference term and the noise term, respectively, and are defined as

$$I(n) := e^{j\frac{2\pi n\epsilon}{N}} \sum_{l \neq 0, l=1}^{L-1} h_l s_u^*(n) s_u((n-l) \bmod N), \quad (10)$$

$$\tilde{v}(n) := s_u^*(n) v(n). \quad (11)$$

The PSS correlation based scheme divides the correlation sequence $\{y(n)\}$ into two equal-length parts, that is $\{y(0), y(1), \dots, y(N/2-1)\}$, $\{y(N/2), y(N/2+1), \dots, y(N-1)\}$. By summing up each divided sequence, we have

$$\begin{aligned} y_{\text{cor},i} &= \sum_{n=Ni/2}^{N(i+1)/2-1} y(n) \\ &= \sum_{n=Ni/2}^{N(i+1)/2-1} e^{j\frac{2\pi n\epsilon}{N}} \frac{h_0 E_s N}{2} + I_{\text{cor},i} + V_{\text{cor},i}, \end{aligned} \quad (12)$$

where $i = 0, 1$, the interference term and the noise term are defined as $I_{\text{cor},i} := \sum_{n=Ni/2}^{N(i+1)/2-1} I(n)$ and

$V_{\text{cor},i} = \sum_{n=Ni/2}^{N(i+1)/2-1} \tilde{v}(n)$, respectively. For the convenience of explaining the CFO estimation idea, we ignore the interference term $I_{\text{cor},i}$ the noise term $V_{\text{cor},i}$, (12) is then rewritten as

$$y_{\text{cor},i} = \sum_{n=Ni/2}^{N(i+1)/2-1} e^{j\frac{2\pi n\epsilon}{N}} \frac{h_0 E_s N}{2}, \quad i = 0, 1. \quad (13)$$

Notice that, the correlation term $y_{\text{cor},2}$ can be written as

$$y_{\text{cor},1} = e^{j\pi\epsilon} y_{\text{cor},0}. \quad (14)$$

Hence, the CFO can be estimated by taking the conjugate multiplication of $y_{\text{cor},0}$ and $y_{\text{cor},1}$, that is

$$\hat{\epsilon}_{\text{cor}} = \frac{\text{angle}(y_{\text{cor},0}^* y_{\text{cor},1})}{\pi}, \quad (15)$$

where $\text{angle}(\cdot)$ is the phase of its argument. Furthermore, over frequency selective fading channels, due to the good CAZAC property of the PSS sequence, the received PSS signals in different paths can be separated, and the final CFO estimation result can be obtained by combining that from each path.

B. The CP Based Scheme

The CP based scheme was originally proposed in [9] by Van de Beek for OFDM based systems, and the algorithm can also be adopted by the 3GPP-LTE systems. Redundant information contained within the CP enables CFO estimation, hence the CP based scheme is bandwidth efficient without transmitting additional pilots. For 3GPP-LTE, define the receiver PSS samples with the CP as $r(n)$, that is

$$r(n) = e^{j\frac{2\pi n\epsilon}{N}} \sum_{l=0}^{L-1} h_l s_u((n-l) \bmod N) + v(n), \quad n = 0, 1, \dots, N + N_g - 1, \quad (16)$$

where $N_g=144$ is the length of CP, and $N=2048$ subcarriers in total for LTE. Only consider the fractional CFO, hence, the CFO can be estimated by

$$\hat{\epsilon}_{\text{CP}} = \frac{\text{angle}(\sum_{k=0}^{N_g-1} r^*(k) r(k+N))}{2\pi}. \quad (17)$$

C. The MUSIC Based Scheme

The MUSIC algorithm was originally proposed in [16] by Schmidt, and in [10], Liu and Tureli proposed a MUSIC based scheme to estimate the CFO for OFDM systems. Consider the matrix-vector representation (4), since $\mathbf{W}_P$ is a subset of the IDFT matrix $\mathbf{W}$, its orthogonal complement is noted as $\mathbf{W}^\perp$ with the element $\mathbf{w}_q$. Define $\mathbf{R} := \{32, 33, \dots, 36, N-36, N-35, \dots, N-32\}$, and $q \in \mathbf{R}$. Clearly, as shown in Fig. 2, the orthogonal vector $\mathbf{w}_q$ corresponds the 10 reserved subcarrier at the ends of the P-SCH. In the absence of the noise and the CFO, i.e., $\epsilon = 0$,

$$\mathbf{w}_q^H \mathbf{r}_{\text{pss}} = \mathbf{w}_q^H \mathbf{W}_P \mathbf{H} \mathbf{d}_u = 0, \quad (18)$$

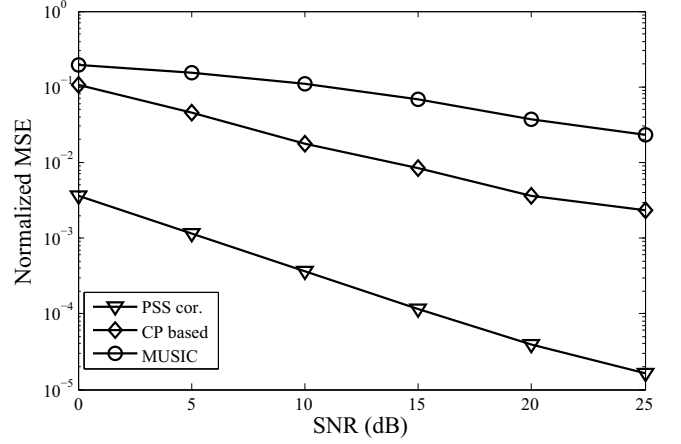


Fig. 3. The performance comparison of the traditional schemes for the 3GPP-LTE downlink over an AWGN channel.

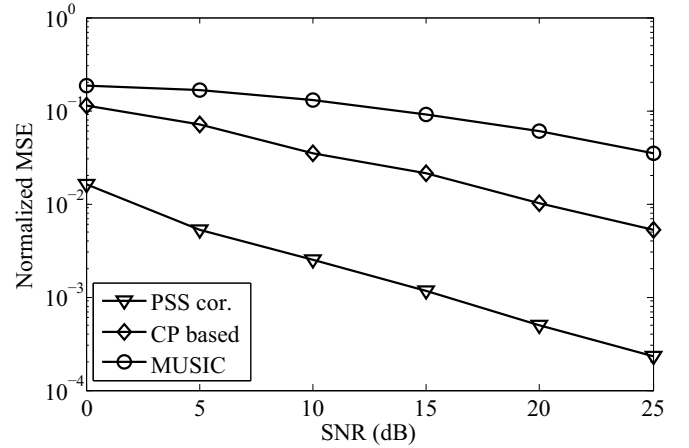


Fig. 4. The performance comparison of the traditional schemes for the 3GPP-LTE downlink over a frequency selective fading channel.

where $q \in \mathbf{R}$. However, this is not true when $\epsilon \neq 0$. Let $\mathbf{Z} = \text{diag}\{1, z, z^2, \dots, z^{N-1}\}$, it can be easily shown that when $\mathbf{Z} = \mathbf{\Gamma}(\epsilon)$,

$$\mathbf{w}_q^H \mathbf{Z}^{-1} \mathbf{r}_{\text{pss}} = \mathbf{w}_q^H \mathbf{Z}^{-1} \mathbf{\Gamma}(\epsilon) \mathbf{W}_P \mathbf{H} \mathbf{d}_u = 0. \quad (19)$$

Consider the noise, a cost function based on the previous observations is as follows:

$$P(z) = \sum_{q \in \mathbf{R}} \|\mathbf{w}_q^H \mathbf{Z}^{-1} \mathbf{r}_{\text{pss}}\|^2 \quad (20)$$

where $z = e^{j\frac{2\pi\epsilon}{N}}$. Therefore, one can find the CFO by evaluating the minimum value of $P(z) = P(e^{j\frac{2\pi\epsilon}{N}})$ along the unit circle, as in the well-known MUSIC [16] algorithm for the array signal processing, that is,

$$\hat{\epsilon}_{\text{MUSIC}} = \arg \min_{\epsilon} P(e^{j\frac{2\pi\epsilon}{N}}). \quad (21)$$

D. Performance Comparison of the Traditional Schemes

To make a comparison of the three reviewed schemes for the downlink of 3GPP-LTE, we evaluate the estimation performances over an AWGN channel and a frequency selective fading channel by using Monte Carlo method.

Fig. 3 and Fig. 4 give the performance comparisons of the three CFO estimation schemes in terms of normalized mean square error (MSE) over an AWGN channel and a frequency selective fading channel, respectively. The frequency selective channel is defined as an 8-ray exponential delay profile uncorrelated Rayleigh fading channel with the time delay between the closest rays being one symbol, and the power of the l^{th} path is

$$E\{\|h_l\|^2\} = \frac{\exp(-\alpha l)}{\sum_{k=0}^{L-1} \exp(-\alpha k)}, \quad l = 0, 1, \dots, L-1, \quad (22)$$

where $\alpha = 1, L = 8$ in the simulation experiments. The normalized MSE is defined as

$$\text{MSE}_\epsilon := \frac{1}{Q} \sum_{q=0}^{Q-1} \|\hat{\epsilon}(q) - \epsilon(q)\|^2, \quad (23)$$

where Q is the number of Monte Carlo runs and $\hat{\epsilon}(q)$ is the CFO estimated at the q^{th} experiment. In the MUSIC based scheme, the step-size in searching the CFO according to (21) is set as $\Delta\epsilon = 0.002$. Clearly, we can observe that the PSS based correlation scheme has better performance compared to the CP based scheme and the MUSIC based scheme. The observations in Fig. 3 and Fig. 4 suggest us that, in order to further improve the performance of CFO estimation, much research attention should be paid to the PSS correlation based scheme for the downlink of 3GPP-LTE. The next section will present an extended CFO estimation scheme of the traditional PSS correlation scheme.

IV. THE PROPOSED CFO ESTIMATION ALGORITHM

In this section, an extended algorithm based on PSS correlation is presented. For convenience, the PSS correlation based scheme reviewed in Section III.A is referred to as the traditional scheme. According to (12)-(15), the CFO estimation is performed by dividing the correlation sequence $\{y(n)\}$ into two parts and calculating their phase difference. Here in our proposed scheme, alternatively, the sequence $\{y(n)\}$ is divided into four parts. Similarly, by calculating the phase difference of any pair among the four parts, we can obtain six CFO estimates. The final CFO estimation value is the linear combination of these six CFO estimates.

In the proposed scheme, the estimator divides the correlation sequence $\{y(n)\}$ into four equal-length parts, that is $\{y(0), y(1), \dots, y(N/4 - 1)\}$, $\{y(N/4), y(N/4 + 1), \dots, y(N/2 - 1)\}$, $\{y(N/2), y(N/2 + 1), \dots, y(3N/4 - 1)\}$, and $\{y(3N/4), y(3N/4 + 1), \dots, y(N - 1)\}$. The respective

sum of each part is written by

$$\begin{aligned} y_{\text{prop.},m} &= \sum_{n=Nm/4}^{N(m+1)/4-1} y(n) \\ &= \sum_{n=Nm/4}^{N(m+1)/4-1} e^{j\frac{2\pi n\epsilon}{N}} \frac{h_0 E_s N}{4} + I_{\text{prop.},m} + V_{\text{prop.},m}, \end{aligned} \quad (24)$$

where $m = 0, 1, 2, 3$, and $I_{\text{prop.},m}$, $V_{\text{prop.},m}$ denote the interference term and the noise term, respectively, that is

$$I_{\text{prop.},m} = \sum_{n=Nm/4}^{N(m+1)/4-1} I(n), \quad (25)$$

$$V_{\text{prop.},m} = \sum_{n=Nm/4}^{N(m+1)/4-1} \tilde{v}(n). \quad (26)$$

By defining $M := \sum_{n=Nm/4}^{N(m+1)/4-1} e^{j\frac{2\pi n\epsilon}{N}} h_0 E_s N/4$ and ignoring $I_{\text{prop.},m}$ and $V_{\text{prop.},m}$, ($m = 0, 1, 2, 3$), for the convenience of explaining the CFO estimation idea, we get that $y_{\text{prop.},0} = M$, $y_{\text{prop.},1} = e^{j\frac{\pi\epsilon}{2}} M$, $y_{\text{prop.},2} = e^{j\pi\epsilon} M$ and $y_{\text{prop.},3} = e^{j\frac{3\pi\epsilon}{2}} M$. Hence, the CFO is estimated by calculating the phase difference of any pair among the four parts. That is

$$\hat{\epsilon}_g = \frac{\text{angle}(y_{\text{prop.},p}^* y_{\text{prop.},q})}{(q-p)\pi/2}, \quad p < q, \quad p, q \in \{0, 1, 2, 3\}, \quad (27)$$

where $g = 1, 2, \dots, 6$. The proposed estimator is the linear combination of $\hat{\epsilon}_g$

$$\hat{\epsilon}_{\text{prop.}} = \sum_{g=1}^6 \omega_g \hat{\epsilon}_g. \quad (28)$$

Since the four parts have equal length and the noises in the four parts have the same property, the simplest way to combine the six estimates in (27) is by setting $\omega_g = 1/6$ in (28).

V. SIMULATION RESULTS

In this section, we compare the proposed CFO estimation scheme with the traditional PSS based estimation scheme. The simulation environment and the parameters are the same as those in Section III.D.

Fig. 5 shows the normalized MSE of the traditional scheme and the proposed one as a function of SNR over the AWGN channel, where the traditional scheme means the PSS correlation based scheme reviewed in Section III.A. The IDFT size for PSS in LTE system is set as 2048. It can be seen from Fig. 5 that compared to the traditional CFO estimator, the proposed CFO estimator has better MSE performance and reduces the required SNR by about 2 dB.

Fig. 6 illustrates the MSE results of different CFO estimation schemes over the frequency selective fading channel whose power delay profile is modeled according to (22). The performance with different values of N , which is the IDFT size for the transformation of PSS sequence in the frequency

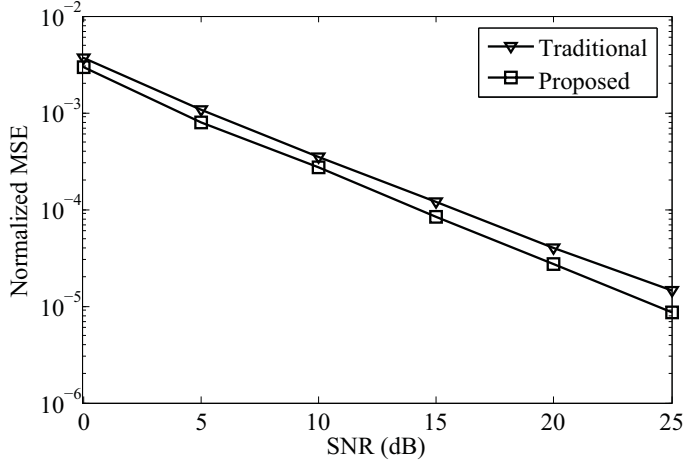


Fig. 5. CFO estimation MSEs of the traditional scheme and the proposed scheme as a function of SNR over the AWGN channel.

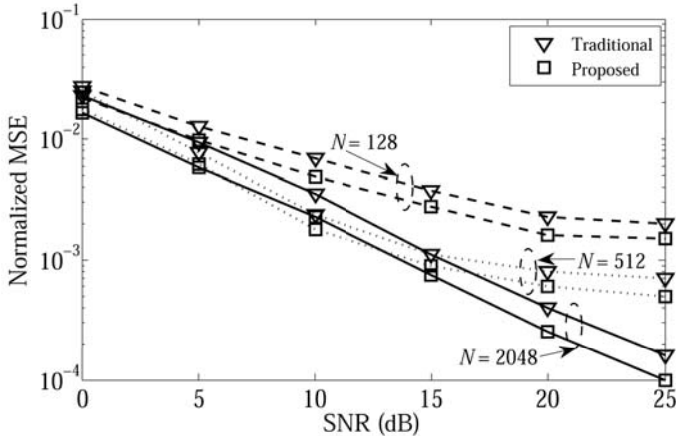


Fig. 6. The normalized MSE performance comparison over the frequency selective fading channel, at different N -point IDFT sizes for PSS sequence in the frequency domain.

domain, is also provided. It can be seen from Fig. 6 that, the larger the IDFT size N is, the smaller MSE both the traditional scheme and the extended scheme achieve. It can also be seen that, the proposed scheme achieves a gain of around 3 dB over the traditional scheme with the same IDFT size.

VI. CONCLUSION

In this paper, we investigated the CFO estimation for 3GPP-LTE downlink systems. Three traditional CFO estimation schemes, i.e, the PSS correlation based scheme, the CP based scheme, and the MUSIC based scheme, have been applied to the downlink of 3GPP-LTE. Furthermore, an extended CFO estimation algorithm based on the PSS correlation scheme has been proposed. Compared to the traditional PSS correlation based scheme, the proposed scheme divides the correlation sequence of PSS into four parts instead of two parts. Simulation results showed that the proposed estimator has better

performance than the traditional one in terms of normalized MSE over both AWGN channels and frequency selective fading channels.

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